Dwell-Adaptive RF Signal Classification from Duration-Complete Synthetic Data

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July 31, 2026

Abstract

The transmitters that matter most—drone control links, intermittent interferers, frequency hoppers—announce themselves for less than a millisecond at a time, while a spectrum monitor can watch only one slice of the band at once: glance briefly and a hopper is indistinguishable from a blip or from nothing; stare long enough to be sure and the rest of the band goes unwatched. Deep I/Q classifiers inherit the worst half of this bargain because they are trained on windows fixed in *samples*, spanning microseconds of millisecond-scale structure; the failure is one of evidence, not modeling, and no architecture can learn rhythm its training windows never contained. We present DACS (Dwell-Adaptive Classification from duration-complete Synthesis), an architecture with a sense of time: one weight-shared, fully-convolutional-in-time network that reads a capture at any dwell, with three heads giving it three faculties—a prototype head that knows the class, a masked denoising autoencoder that knows the channel (its target the true clean signal, silences included), and a confidence head that knows itself—so that listening longer becomes an action the classifier takes, capture by capture, rather than a compromise fixed at design time. The capability is learnable only from data in which time is a controlled variable: synthetic captures long enough to hold each protocol's timing, silences retained as class-conditional evidence, each paired with its clean reference. With architecture, optimizer, and budget frozen, duration-complete training alone lifts 7-class balanced accuracy from 0.503 to 0.954 at 1 ms dwell and 1.000 at 10 ms across two seeds, while architecture substitution, tripled budget, and worst-case reweighting on fixed-window data all fail; a companion analysis shows why standard worst-cell metrics could not even have displayed the win. Training cost is linear in dwell.

Index terms—RF signal classification, selective classification, anytime inference, synthetic training data, frequency hopping.

1 Introduction

The air around every airport fence, factory floor, and city block carries transmitters no one can see—and the ones that matter most, the drone control link, the interferer that kills a production line three packets at a time, announce themselves for less than a millisecond before hopping somewhere else. A spectrum monitor can listen to only one slice of that air at a time, so it faces an impossible bargain: glance briefly and a frequency hopper is indistinguishable from a harmless blip—or from nothing at all—while staring long enough to be certain leaves the rest of the band unwatched. Deep classifiers [1, 2] should resolve this bargain, but trained on fixed-length windows that span microseconds of millisecond-scale structure, they inherit its worst half: they answer instantly, and they answer wrong, on exactly the signals they exist to catch.

This failure looks like a modeling problem and has been attacked as one—bigger networks, longer training, cleverer losses—but it is an evidence problem: burst protocols are defined by rhythms measured in milliseconds, GSM's 4.6 ms frame [14], Bluetooth's 625 μs slot cadence [13], while fixed-sample-count training windows span microseconds of that rhythm, and no architecture, budget, or loss function can recover structure its training windows physically never contained (Fig. 1). Radio engineering's founding instinct is to separate the signal from the noise—but burst protocols turn the instinct against us: the silence between bursts is not noise to be rejected, it is the signal's own rhythm, and every pipeline that discards "empty" windows as invalid is, quite literally, separating the signal from itself—because for spectro-temporal signals, the signal is sometimes the noise.

Since no fixed dwell can win—short looks miss identity, long looks miss the band—the classifier itself must decide how long to listen. DACS (Dwell-Adaptive Classification from duration-complete Synthesis) is built around that decision: one weight-shared network reads a capture at any dwell, classifies it against prototypes, reconstructs what the clean signal should have looked like (so silence is knowledge, not absence), and scores its own reliability—escalating to a longer look only when that score demands it. Training this sense of time requires data in which time is a controlled variable: captures long enough to hold each protocol's timing, with silence retained at its true class-conditional rate. Sec. 3 verifies the diagnosis quantitatively—including a third, less obvious casualty: the worst-cell acceptance metrics of this domain are order statistics that could not have displayed success even if a model achieved it. Secs. 4–5 give the classifier and the data technique; Sec. 6 then tests the diagnosis clause by clause: swapping only the data moves balanced accuracy from 0.503 to 0.954 at 1 ms dwell (1.000 at 10 ms, two seeds), while every model-side lever—architecture, budget, loss—fails to approach it.

2 Related Work

Deep I/Q classification varies architectures under a fixed observation contract [1, 2]; DACS changes the contract and supplies the architecture built to exploit it. Its components are established—prototype heads [3], denoising and masked autoencoders [5, 4], trained confidence [6, 7]—but their composition realizes selective prediction [8] and early exiting [10, 11] along a new axis: physical observation time rather than depth, with calibration [9] governing escalation.

3 Problem Analysis

Each claim of the diagnosis is checkable. We check them on the *fixed-count control corpus*: captures of 16,384 samples at rates swept over 1–200 Msps (median TDMA-class window 1.55 ms), constructed to instantiate the prevailing convention with the same classes and impairment envelope as our method's corpus; we call such data *duration-starved*. Throughout, *worst-window balanced accuracy* denotes the minimum over the corpus's three window tiers (4,096/8,192/16,384 samples) of 7-class balanced accuracy.

The windows really are empty. Against the stationary expectation that the first quarter of a capture holds 25% of its energy, 37% of GSM captures hold <10% of capture energy in their first 4,096 samples (Fig. 2a aggregates all GSM configurations; burst-mode configurations individually range 40–48%, the always-on broadcast carrier ≈0%; block crest factor 39×); frequency-hopped captures 23%; DSSS 24%; continuous classes ≈0%. Conventional pipelines reject these windows as invalid—discarding $P(\text{silence} \mid \text{class})$, the statistic that a prior-optimal analysis shows is alone worth balanced accuracy near 0.93.

Energy is not the limit; rhythm is. The continuously transmitting GSM configuration (loaded broadcast carrier)—never empty—still yields only 0.54 recall on the shortest window tier under the feature-engineered baseline of Sec. 6 (mean over its 24 tuned configurations). The GSM/Bluetooth discriminant is frame timing versus hop timing, both invisible inside a sub-millisecond span, exactly as the diagnosis predicts.

Success could not even be displayed. Acceptance criteria for deployed classifiers often take the form of minimum recall over (configuration×condition) cells. With J cells of n evaluation rows the displayed value is $\mathbb{E}[\min_j \hat{p}_j]$, $\hat{p}_j \sim \text{Bin}(n, p)/n$; for a representative design with J=124 cells of n=16 (e.g., 31 configurations crossed with 4 conditions), true per-cell recall 0.95/0.97/0.99 reads as 0.77/0.82/0.89 (Fig. 2b): a solved system reports 0.8. The remedy is adequate cells, not a different statistic: at J=34, n=96, a true 0.97 displays as 0.93. We adopt per-configuration cells of n=96 per dwell throughout, reporting balanced accuracy alongside the minimum cell.

4 Method

The escape from the scanner's bargain is a network that treats dwell as a decision variable—an instrument with three faculties: it knows the class, it knows the channel, and it knows itself. A single spectro-temporal encoder (STFT front end; six 3×3 convolution–GroupNorm–SiLU blocks, fully convolutional in time, 32× temporal downsampling; 1.0M parameters alone, 1.57M with the decoder and heads below) feeds three heads (Fig. 4). With episode windows x at dwells drawn from an ordered set $\ell \in \{\ell_1 < \cdots < \ell_K\}$ (we use $\{1, 2.5, 10\}$ ms), spectrogram S_x, random time-frame mask m, and class prototypes computed from support embeddings [3]:

$$\mathcal{L} = \mathcal{L}_{\text{proto}} + \lambda_r \left\| D\big(E(m \odot S_x)\big) - S_{\text{clean}} \right\|_1 + \lambda_d(t)\, \mathbb{E}_i \Big[\max_{\ell} \mathcal{L}_{\text{proto}}\big(x_i^{(\ell)}\big) \Big] + \lambda_c\, \text{BCE}\big(g(x), \mathbb{1}[\hat{y} = y]\big).$$

$\mathcal{L}_{\text{proto}}$ is episodic cross-entropy over squared prototype distances with a learned temperature. The second term is a masked denoising autoencoder [5, 4] with exact supervision: $D \circ E$ must reproduce the capture's true clean reference—not a proxy corruption of the input—so silence is a target rather than an exclusion: for an idle stretch of a burst protocol the correct reconstruction is the noise floor, and the network is graded on knowing where the rhythm's rests fall (Fig. 3 breaks this supervision down on a real capture). Pseudocode for the episode loss is given in Algorithm 1. The third term trains the shortest dwell against its hardest cases (max over dwells of the same rows' episodic loss; prototypes detached; weight ramped linearly from zero). The fourth trains g to predict the classifier's own correctness [6, 7]. Optimization uses AdamW with cosine annealing [12].

Algorithm 1 DACS training-episode loss. N_C=7 classes per episode; N_S, N_Q support/query captures per class; dwell set $\{\ell_1 < \ldots < \ell_K\}$; $\lambda_r, \lambda_d(t), \lambda_c$ loss weights (λ_d ramped). RandomSample(S, N) draws N elements uniformly without replacement; Window(x, ℓ) extracts a random-offset window of duration ℓ from capture x; Sp is the STFT stack; m a random time-frame mask. $\mathcal{D}_k$ is the training subset with class k; every capture x is paired with its clean reference $\tilde{x}$.

Input: training set $\mathcal{D}$ with clean pairs; encoder E, decoder D, embedding f_ϕ, confidence head g
Output: episode loss J
$\ell \leftarrow \textsc{RandomSample}(\{\ell_1, \ldots, \ell_K\}, 1)$ ▹ Draw episode dwell
for k in $\{1, \ldots, N_C\}$ **do**
  $S_k \leftarrow \textsc{RandomSample}(\mathcal{D}_k, N_S)$; $Q_k \leftarrow \textsc{RandomSample}(\mathcal{D}_k \setminus S_k, N_Q)$ ▹ Support/query captures
  $x \leftarrow \textsc{Window}(x, \ell)$ for every capture drawn ▹ Empties retained
  $\mathbf{c}_k \leftarrow \frac{1}{N_S} \sum_{x \in S_k} f_\phi(\text{Sp}(x))$ ▹ Class prototype
end for
$J \leftarrow \sum_k \sum_{x \in Q_k} \frac{1}{N_C N_Q} \text{CE}\big(-d(f_\phi(\text{Sp}(x)), \mathbf{c}.)\big)$ ▹ Know the class
$J \leftarrow J + \lambda_r \| D(E(m \odot \text{Sp}(x))) - \text{Sp}(\tilde{x}) \|_1$ ▹ Know the channel: exact target
$J \leftarrow J + \lambda_d(t)\, \mathbb{E}_i \big[\max_{\ell'} \text{CE}_i^{(\ell')} \big]$ ▹ Worst dwell; amortized
$J \leftarrow J + \lambda_c\, \text{BCE}\big(g(x), \mathbb{1}[\hat{y} = y]\big)$ ▹ Know thyself

Dwell policy. At inference the system classifies at ℓ_1 and consults g: if calibrated confidence [9] clears a threshold the answer is emitted; otherwise the capture extends to the next dwell—selective classification [8] in which the cost of abstention is observation time, and the analogue of early exiting [10, 11] with physical time replacing depth. Algorithm 2 states the policy; Fig. 5 shows its payoff against observation time. The monitor spends milliseconds only on the captures that demand them; everywhere else, the band stays watched.

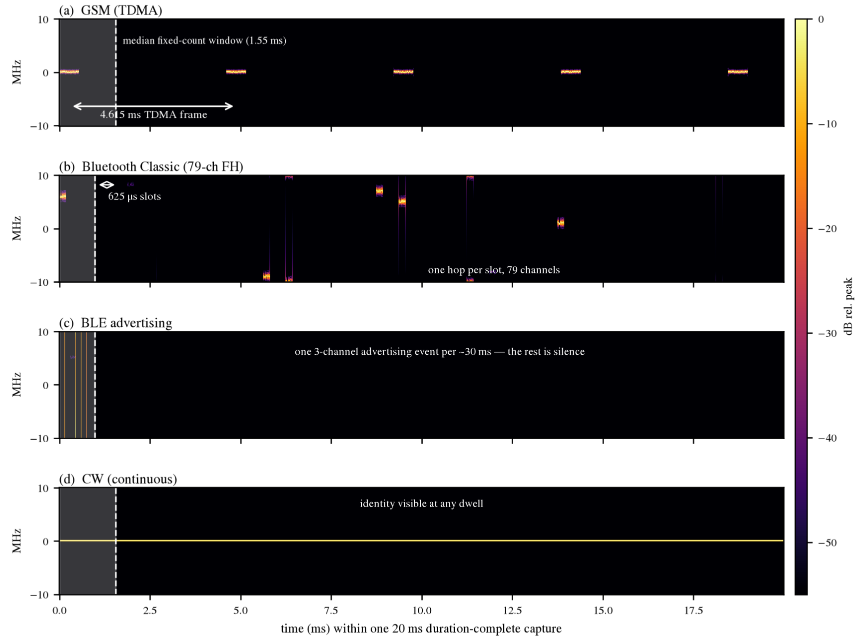

Figure 1: Class identity lives at timescales a fixed-sample-count window rarely contains. Spectrograms of single 20 ms duration-complete captures (clean synthesis, 20 Msps): (a) GSM, (b) Bluetooth Classic, (c) BLE advertising, (d) CW. Shading marks the median fixed-count window (16,384 samples, 1–200 Msps sweep): within it, GSM shows at most one burst, Classic at most one hop, BLE one event or pure silence. The 20 ms capture contains the discriminants: TDMA framing [14], slot-clocked hopping and sparse advertising [13], continuity for CW.

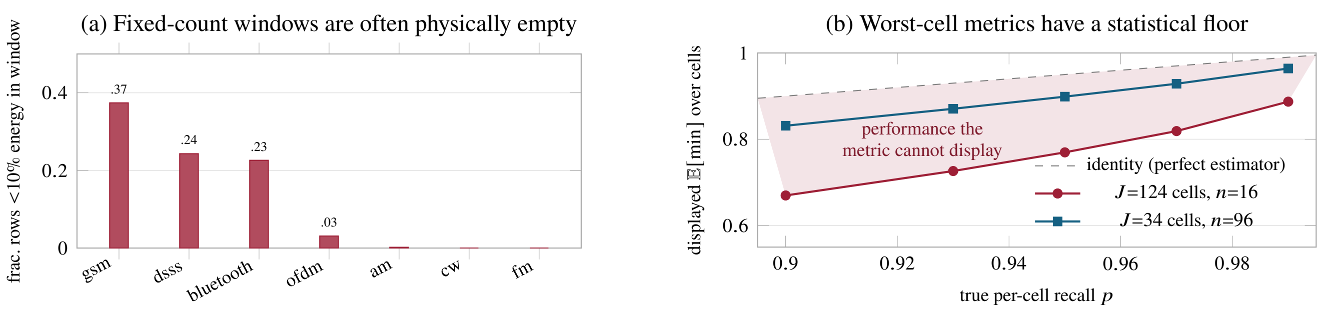

Figure 2: The premises, measured. (a) Fraction of fixed-count corpus rows whose scored window contains <10% of capture energy: the hardest classes are frequently *not in the window*. (b) Monte Carlo readout ($2{\times}10^4$ draws) of minimum-over-cells metrics: small-n cells cap what any model can *display*, independent of its true performance.

Algorithm 2 Dwell-adaptive inference. τ: confidence threshold calibrated on held-out data; prototypes $\{\mathbf{c}_k\}$ from enrollment; a capture keeps recording while the loop runs.

Input: capture stream x; dwell set $\{\ell_1 < \ldots < \ell_K\}$; threshold τ
Output: label $\hat{y}$, dwell spent
  for j in $\{1, \ldots, K\}$ **do**
    $z, g \leftarrow f_\phi(\mathrm{Sp}(x[0{:}\ell_j])),\ g(x[0{:}\ell_j])$ ▷ Re-encode the grown capture
    $\hat{y} \leftarrow \arg\min_k d(z, \mathbf{c}_k)$ ▷ Prototype decision
    if $g \geq \tau$ **or** $j = K$ **then**
      return $(\hat{y}, \ell_j)$ ▷ Answer, or forced at max dwell
    end if
  end for ▷ Otherwise: keep listening

5 Dataset Generation

Training requires corpora whose captures contain the structure of Fig. 1; we summarize the technique and refer details to the artifact. Every generator is *index-pure*—a deterministic function $y[n] = G(n)$ of absolute sample index—so arbitrary-duration captures assemble byte-exactly from bounded, validated packet content. Burst processes replay standards-derived packets [13] on unbounded timelines (slot-clocked keyed-hash hopping for Bluetooth Classic; an event grid with the specification's *advDelay* for BLE), reproducing occupancy and timing *statistics*; the hop-selection kernel and payload variation are out of scope. Clean captures are synthesized at native rates, resampled to 20 Msps, and impaired by a seeded receiver chain (AWGN at $U(-5, 45)$ dB SNR on active-sample power, carrier offset, phase noise, IQ imbalance, DC, multipath), yielding the row-aligned clean/impaired pairs of Fig. 3. Training windows are drawn by duration at random offsets; empty windows are retained.

6 Experiments and Results

The diagnosis of Sec. 1 makes falsifiable predictions: swapping the data should solve the problem; improving the model should

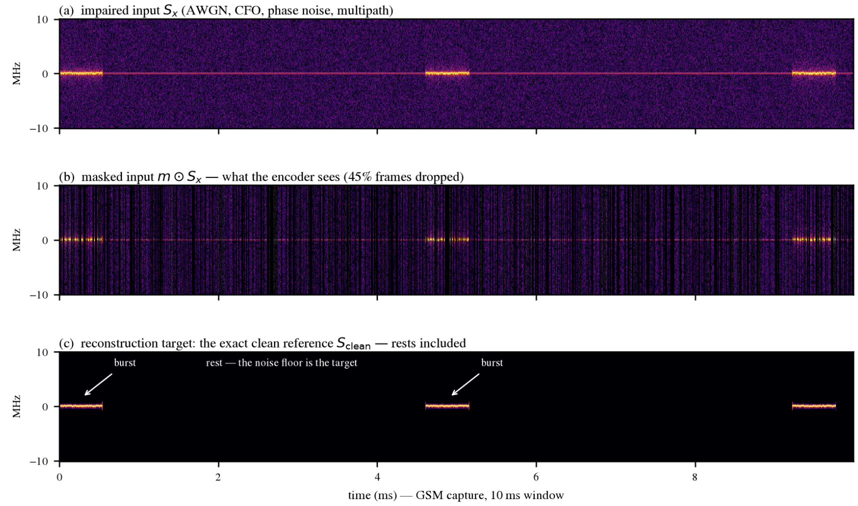

Figure 3: Exact-target denoising supervision, on a real GSM capture. (a) The impaired input: three TDMA bursts under AWGN, carrier offset, phase noise, and multipath. (b) The masked view the encoder receives. (c) The reconstruction target is the *true clean reference* from paired synthesis—not a corruption of the input—so the rests between bursts are graded targets: the network must know where the rhythm's silences fall. This supervision is unavailable to classic denoising autoencoders and to any over-the-air dataset.

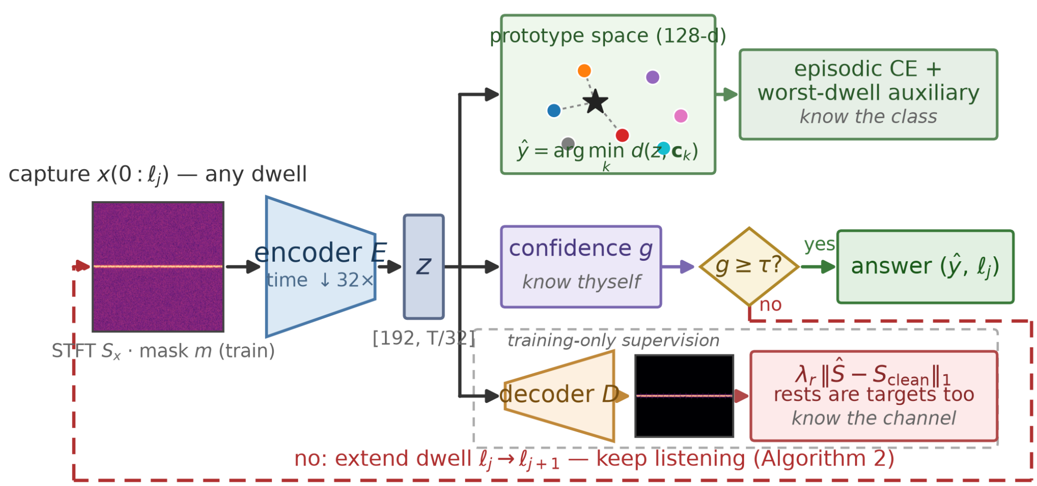

Figure 4: The DACS network. One shared spectro-temporal encoder; three heads. The prototype embedding [3] carries classification and preserves enrollment-based open-set operation; the confidence head [6, 8] drives the answer-early/escalate dwell policy; the light decoder reconstructs the clean reference's spectrogram under time-frame masking [4, 5], making silence a learning target and regularizing against the shortcut overfit that duration-starved training exhibits (Sec. 6). The worst-case-over-dwell auxiliary trains the shortest dwell against its hardest cases.

not. We test both.

Setup. Seven classes (`am`, `bluetooth`, `cw`, `dsss`, `fm`, `gsm`, `ofdm`) spanning 34 waveform configurations; 8,704 captures of 20 ms, 96 evaluation rows per configuration (160 further captures per configuration for training); impairments as in Sec. 5. The fixed-count control corpus is that of Sec. 3. Training budget denotes the number of episodic updates (7-way, 5-shot/5-query; dwell drawn uniformly per episode); the base budget is 8,000 episodes and "3×" is 24,000. Optimization: AdamW [12], cosine annealing, peak learning rate 2×10^{-3}; loss weights λ_r=0.3, λ_d=0.15 (linear ramp over the first tenth of training), λ_c=0.1. Evaluation classifies the capture-initial window of each evaluation row at each dwell, with class prototypes computed from 64 training captures per class under the identical protocol on both corpora.

Prediction 1: the data solves it. Holding the encoder (1.0M parameters, prototype loss only—no reconstruction, worst-case, or confidence terms), optimizer, and budget fixed and swapping only the corpus (Fig. 5): balanced accuracy rises from 0.503 on the shortest fixed-count tier to 0.954 at 1 ms dwell, and from 0.682 to 1.000 at the longest tier vs. 10 ms, with every class including GSM and Bluetooth at recall 1.000 at 10 ms; an independent seed reproduces 0.958/0.972/1.000. Criteria fixed before the runs executed (GSM and BT ≥0.85 at 10 ms; short-dwell balanced ≥ the control level; monotone recall in dwell) all passed on both seeds. Because the control corpus sweeps

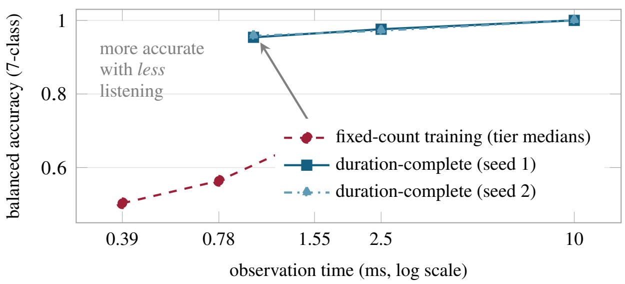

Figure 5: Accuracy vs. observation time. The identical 1.0M-parameter encoder trained on fixed-count data (tiers at their median physical durations) and on duration-complete data (two seeds): more accurate at every operating point with *less* listening; at 10 ms every class reaches recall 1.000 on both seeds.

Table 1: Controls on the fixed-count corpus: no model-side intervention approaches the effect of duration-complete data.

Intervention (control corpus)	Metric	Effect
Feature-eng. baseline, 50-trial search	worst-win. bal.	0.743
its GSM recall, shortest tier	GSM recall	0.49
Our encoder, raw / normalized input	GSM recall	0.34 / 0.45
3× training budget	worst-win. bal.	0.681 → 0.657
Worst-case loss, wt. .05/.15/.40	Δ w.-win.	+.003/+.009/−.008
Worst-case loss, 3× budget	Δ w.-win.	+.055
Duration-complete data, fixed encoder	shortest-tier bal.	0.503 → 0.954

sample rate, its tiers are matched to dwells by rank (shortest/mid/longest), not by physical duration, which fixed-count data cannot hold constant.

Prediction 2: the model does not (Table 1). On the fixed-count control corpus: (i) the same spectro-temporal encoder *loses* to a feature-engineered baseline—hand-designed invariant patch preprocessing (sixteen normalized 64-sample patches with geometry canonicalization) feeding a compact CNN, tuned by a 50-trial hyperparameter search—scoring 0.34 (raw spectrogram input) and 0.45 (adding per-window RMS normalization and phase augmentation) vs. the baseline's 0.49 GSM recall on the shortest tier. (ii) Tripling the training budget *reduces* worst-window balanced accuracy, 0.681→0.657, the signature of shortcut overfitting on ambiguous windows. (iii) A worst-case-over-window auxiliary (max cross-entropy over window tiers) peaks at +0.009 over the 0.681 base-budget baseline (inverted-U dose response, weight ≈0.15) and recovers +0.055 at 3× budget (0.712 vs. 0.657). Architecture, budget, and loss each fail exactly as an evidence-limited diagnosis requires.

Full system. At an early checkpoint one third through its training schedule, the DACS network reaches balanced accuracy 0.909/0.950/0.990 at 1/2.5/10 ms with minimum-cell recall 0.448/0.760/0.948; surviving errors concentrate in single configurations at the shortest dwell—precisely the cases escalation targets. The raw confidence head predicts the answer-at-1 ms decision at 0.884 accuracy against a 0.909 marginal correctness rate at this checkpoint; the head is uncalibrated here, and we accordingly claim the escalation *mechanism*, deferring its operating curve to calibration on held-out data (Sec. 7). Training cost is linear in dwell: 1.4 ms (1 ms window) to 31 ms (20 ms window) per training step per window on one Apple M-series GPU, so duration completeness is computationally cheap.

7 Limitations

All data are synthetic; over-the-air validation remains future work [2]. The hop composition reproduces occupancy and timing statistics, not the specification's selection kernel, and repeats verified payload bits. Captures contain a single emitter; co-channel interference and Doppler are absent. Confidence calibration [9] on held-out data is required before escalation thresholds are meaningful, and the full-system numbers above are an early snapshot at one third of the training schedule.

8 Conclusion

The scanner's bargain—glance and be fooled, stare and go blind—was never going to be resolved by a better glance. It is resolved by an architecture that can spend observation time deliberately—classifying, modeling the clean channel, and judging its own reliability at every dwell—and by training data whose windows contain the rhythms that make that judgment learnable. With the model frozen, duration-complete data moved the problem from 0.503 to 0.954 at 1 ms; with the data frozen, no model-side intervention came close. When a worst-case metric plateaus across model families, audit what the observation window physically contains—and what the metric can statistically display—before spending another epoch on the model. Sometimes the signal is the noise—and sometimes the answer is to keep listening.

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